



View all journals

Search

Log in

Explore content About the journal Publish with us Subscribe

Sign up for alerts

RSS feed

nature > news > article

NEWS | 01 November 2024

Chemists make ‘impossible’ molecules that break 100-year-old bonding rule

The molecules were deemed too unstable to make, but experiments show they can be ‘captured’ in certain reactions to yield useful products.

By [Gemma Conroy](#)



Coloured scanning electron micrograph of Soot (carbon) particles from the inside of a wood burning chimney.

Organic molecules, which contain carbon (pictured), form certain shapes because of how their atoms bond.Credit: Dennis Kunkel Microscopy/SPL

For the first time, [chemists](#) have made a class of molecules previously thought too unstable to exist, and used them to generate exotic compounds¹. Scientists say these notorious molecules, known as anti-Bredt olefins (ABOs), offer a new path to synthesizing challenging drug candidates.

The work is “a landmark contribution”, says Craig Williams, a chemist at the University of Queensland in Brisbane, Australia. The findings are published in *Science*.

Organic molecules, which contain [carbon](#), typically conform to specific shapes because of how their atoms bond together. For instance, olefins, also called alkenes — hydrocarbons that are often used in reactions for [drug development](#) — have one or more double bonds between two carbon atoms, resulting in the atoms being arranged in one plane.

The 100-year-old Bredt’s rule — which was proposed in 1924 by [organic chemist](#) Julius Bredt — states that, in small molecules made up of two rings that share atoms, such as some types of alkene, double bonds between two carbon atoms cannot occur where the rings join together, called the bridgehead position. This is because the bonds would force the molecule into a tortured, strained 3D shape that makes it highly reactive and unstable, says study co-author Neil Garg, a chemist at the University of California, Los Angeles. “Still, 100 years later, people would say these types of structures are forbidden or too unstable to make,” he says.

Although the rule has made its way into chemistry textbooks, it hasn’t stopped researchers from trying to break it. Previous research has hinted that it’s possible to create ABOs that have a double carbon–carbon bond at the bridgehead position². But attempts to synthesize them in their complete form were unsuccessful because the reaction conditions were too harsh, says Garg.

Trapping agents

In the latest attempt, Garg and his colleagues treated a precursor compound with a fluoride source to kick-start a milder ‘elimination’ reaction, which removes groups of atoms from molecules. This resulted in a molecule that had the telltale ABO double-carbon bond. When the researchers added various trapping agents — chemicals that capture unstable molecules as they react — to this 3D ABO, they were able to produce several complex compounds that could be isolated. This suggests that the reactions of ABOs with different trapping agents can be leveraged to synthesize 3D molecules, which are useful for designing new drugs, says Garg.

Unlike typical alkenes, ABOs are chiral compounds — molecules that don’t perfectly match up with their mirror image. Garg and his colleagues synthesized and trapped an ABO that was enantioenriched, which means they produced more of one mirroring pair than the other. This finding suggests that ABOs could be used as unconventional building blocks for enantioenriched compounds, which are widely used in pharmaceuticals.

Chuang-Chuang Li, a chemist at the Southern University of Science and Technology in Shenzhen, China, says that the approach could be used to explore innovative synthesis pathways for other challenging molecules, such as the chemotherapy drug paclitaxal (sold as Taxol) — a complex, multi-ringed molecule that is difficult to create in the laboratory. “It’s a valuable and reliable method,” says Li.

Garg and his team are exploring other reactions involving ABOs, and are investigating how to synthesize other molecules with seemingly impossible structures. “We can be thinking a little bit more outside of the box,” he says.

doi: <https://doi.org/10.1038/d41586-024-03538-4>

References

- McDermott, L. *et al. Science* **386**, eadq3519 (2024).
[Article](#) [Google Scholar](#)
- Chan, T. H. & Massuda, D. *J. Am. Chem. Soc* **99**, 936–937(1977).
[Article](#) [Google Scholar](#)

[Download references](#) ↓

[Reprints and permissions](#)

Latest on:

- [Organic chemistry](#)
 - [Chemistry](#)
 - [Drug discovery](#)
- [Direct conversion from alkenes to alkynes](#)
[Article 16 MAR 26](#)
 - [Alcohol group migration by proximity-enhanced H atom abstraction](#)
[Article 10 MAR 26](#)
 - [Markovnikov hydroamination of terminal alkenes via phosphine redox catalysis](#)
[Article 23 FEB 26](#)

Nature Careers

Jobs >

- [SENIOR OR MID-CAREER RESEARCHERS IN AI AND DATA SCIENCE AND COMPUTATIONAL SCIENCES \(F/M\)](#)
The Paris Brain Institute invites expressions of interest from internationally recognized senior and mid-career researchers.
Paris, Ile-de-France (FR)
Institut du Cerveau
- [Full Professorship \(W3\) for „Clinical Epidemiology of Early Cancer Detection“ \(f/m/d\)](#)
Full Professorship (W3) for ‚Clinical Epidemiology of Early Cancer Detection‘ (f/m/d) We are looking for a highly distinguished physician scientist...
Heidelberg, Baden-Württemberg (DE)
Medizinische Fakultät Mannheim der Universität Heidelberg
- [Associate Editor / Senior Editor \(Discover Journals\)](#)
Job Title: Associate Editor / Senior Editor (Discover Journals) Location: Nanjing (Primary), Beijing, Shanghai, or Pune- hybrid working model Ab...
Nanjing (Primary), Beijing, Shanghai, or Pune- hybrid working model
Springer Nature Ltd
- [Seeking Talents: Faculty Positions in AI at Institute of Automation, Chinese Academy of Sciences](#)
Full/Associate/Assistant Professor Positions of State Key Labs are opening.
Beijing, Nanjing and Wuhan
Institute of Automation, Chinese Academy of Sciences
- [Associate or Senior Editor. BMC Digital Health. BMC Artificial Intelligence, BMC Biomedical Engineer](#)
This is a dynamic and varied role that will give you the opportunity to see content across a broad range of topics.
Pune or Shanghai – hybrid working model
Springer Nature Ltd

Nature (Nature) | ISSN 1476-4687 (online) | ISSN 0028-0836 (print)

About Nature Portfolio

- [About us](#)
- [Press releases](#)
- [Press office](#)
- [Contact us](#)

Discover content

- [Journals A-Z](#)
- [Articles by subject](#)
- [protocols.io](#)
- [Nature Index](#)

Publishing policies

- [Nature portfolio policies](#)
- [Open access](#)

Author & Researcher services

- [Reprints & permissions](#)
- [Research data](#)
- [Language editing](#)
- [Scientific editing](#)
- [Nature Masterclasses](#)
- [Research Solutions](#)

Libraries & institutions

- [Librarian service & tools](#)
- [Librarian portal](#)
- [Open research](#)
- [Recommend to library](#)

Advertising & partnerships

- [Advertising](#)
- [Partnerships & Services](#)
- [Media kits](#)
- [Branded content](#)

Professional development

- [Nature Awards](#)
- [Nature Careers](#)
- [Nature Conferences](#)

Regional websites

- [Nature Africa](#)
- [Nature China](#)
- [Nature India](#)
- [Nature Japan](#)
- [Nature Middle East](#)

- [Privacy Policy](#)
- [Use of cookies](#)
- [Your privacy choices/Manage cookies](#)
- [Legal notice](#)
- [Accessibility statement](#)
- [Terms & Conditions](#)
- [Your US state privacy rights](#)

You have full access to this article via **Bodleian Libraries of the University of Oxford**.

Related Articles

[‘Almost magical’: chemists can now move single atoms in and out of a molecule’s core](#)

[Revolutionary microscopy technique sees individual atoms for first time](#)

[Most accurate clock in history made by ‘quieting’ atoms](#)

[Chemistry Nobel goes to developers of AlphaFold AI that predicts protein structures](#)

[How the periodic table survived a war to secure chemistry’s future](#)

[Chemistry can help make plastics sustainable — but it isn’t the whole solution](#)

[Chemistry is inaccessible: how to reduce barriers for disabled scientists](#)

Subjects

[Organic chemistry](#) [Chemistry](#) [Drug discovery](#)

Sign up to Nature Briefing

An essential round-up of science news, opinion and analysis, delivered to your inbox every weekday.

Email address

☐ Yes! Sign me up to receive the daily *Nature Briefing* email. I agree my information will be processed in accordance with the *Nature* and Springer Nature Limited [Privacy Policy](#).

